Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A full-wave rectifier is fed with ac mains of frequency $50\ \text{Hz}$. What is the fundamental frequency of the ripple in the output current?

(A) $25\ \text{Hz}$

(B) $50\ \text{Hz}$

(C) $75\ \text{Hz}$

(D) $100\ \text{Hz}$

Q.2

Ripple frequency at the output of bridge rectifier when the transformer primary is connected to A.C. mains supply is

(A) $50\ \text{Hz}$

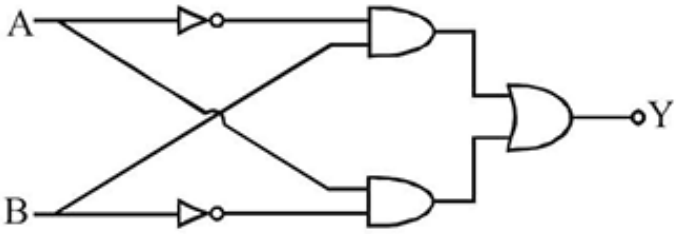
(B) $100\ \text{Hz}$

(C) $25\ \text{Hz}$

(D) $75\ \text{Hz}$

Q.3

The following circut diagram represents



(A) OR gate

(B) XOR gate

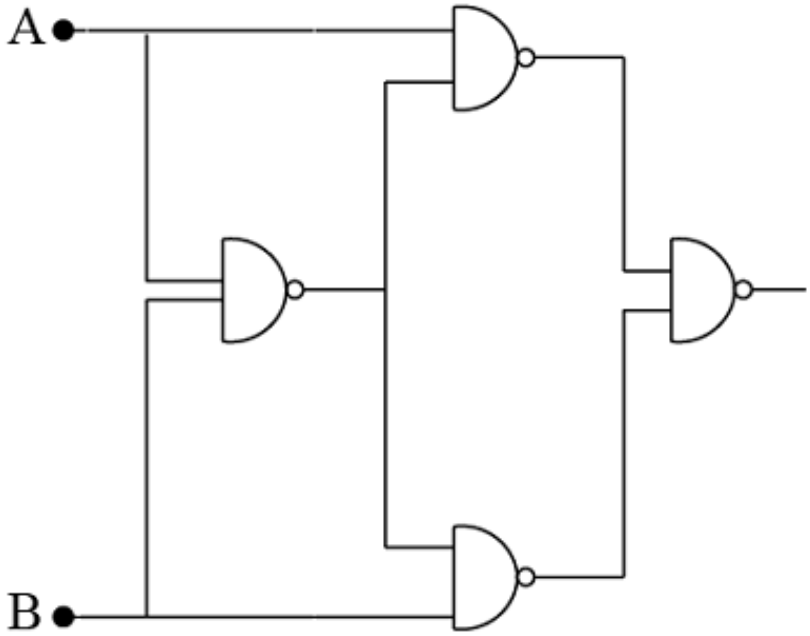
(C) AND gate

(D) NAND gate

Q.4

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The correct truth table for system of four NAND gates as shown in figure is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

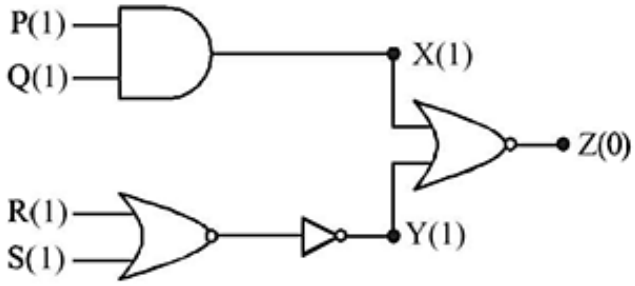
(D)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

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Q.5

The circuit diagram shows a logic combination with the states of outputs X, Y and Z given for inputs $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ and S all at state 1 . When inputs P and R change to state 0 with inputs Q and S still at 1, the states of outputs X, Y and Z change to



(A) $1, 0, 0$

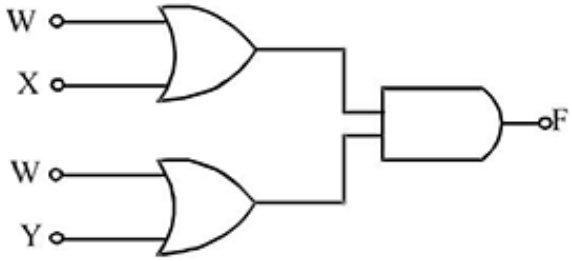
(B) $1, 1, 1$

(C) $0, 1, 0$

(D) $0, 0, 1$

Q.6

The diagram of a logic circuit is given below. The output F of the circuit is represented by



(A) $\mathrm{W} \cdot (\mathrm{X} + \mathrm{Y})$

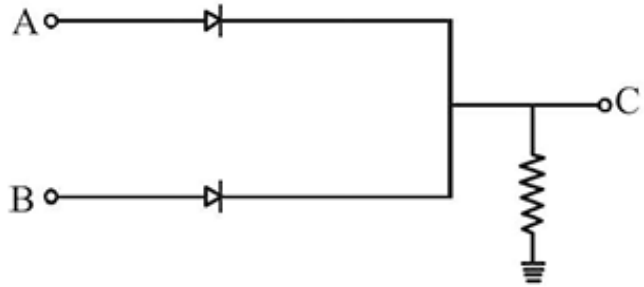
(B) W. (X. Y)

(C) $\mathrm{W} + (\mathrm{X}.\mathrm{Y})$

(D) $\mathrm{W} + (\mathrm{X} + \mathrm{Y})$

Q.7

In the circuit given below, A and B represent two inputs and C represents the output.
The circuit represents



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(A) NOR gate

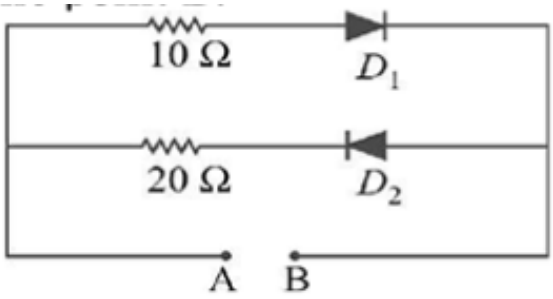
(B) AND gate

(C) NAND gate

(D) OR gate

Q.8

A $5\ \mathrm{V}$ battery may be connected across the points
A and B as shown in figure. Assume that the resistance of each diode is zero in forward bias and infinity in reverse bias. Find the current i_2 by the battery if the positive terminal of the battery is connected to the point B.



(A) $\mathrm{i}_2 = 1.3\ \mathrm{A}$

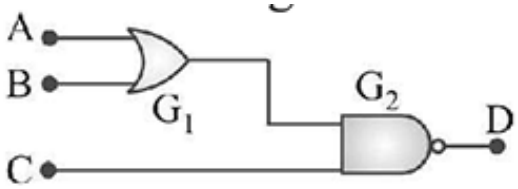
(B) $\mathrm{i}_2 = 2.8\ \mathrm{A}$

(C) $\mathrm{i}_2 = 0.25\ \mathrm{A}$

(D) $\mathrm{i}_2 = 0.75\ \mathrm{A}$

Q.9

For the given combination of gates, if the logic states of inputs A, B, C are as follows $A = B = C = 0$ and $\mathrm{A} = \mathrm{B} = 1, \mathrm{C} = 0$ then the logic states of output D are



(A) 0,0

(B) 0,1

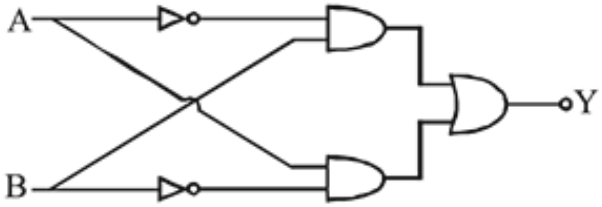
(C) 1,0

(D) 1,1

Q.10

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The following circut represents



(A) OR gate

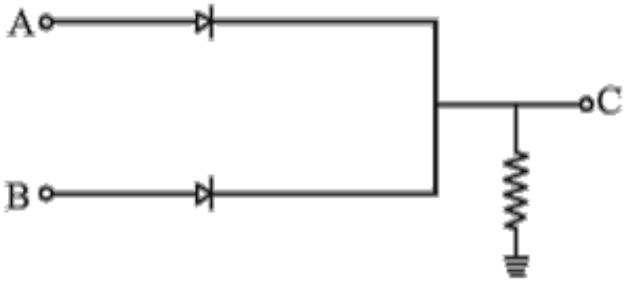
(B) XOR gate

(C) AND gate

(D) NAND gate

Q.11

In the circuit below, A and B represent two inputs and C represents the output. The circuit represents



(A) NOR gate

(B) AND gate

(C) NAND gate

(D) OR gate

Q.12

What will be the input of A and B for the Boolean expression $\overline{(A+B)' \cdot (A \cdot B)'} = 1$

(A) 0,0

(B) 0,1

(C) 1,0

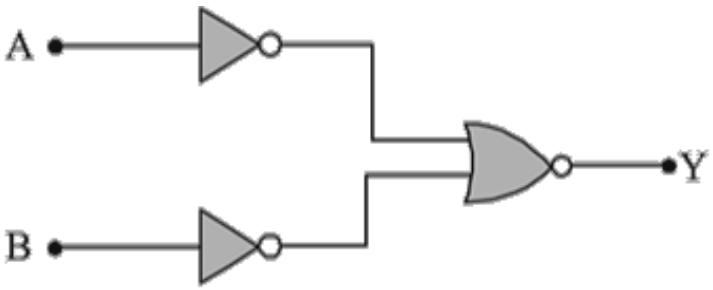
(D) 1,1

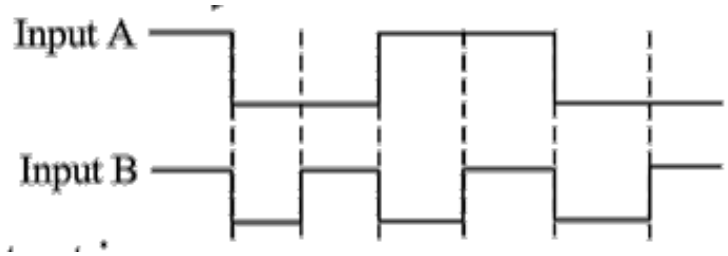
Q.13

The logic circuit shown below has the input waveforms 'A' and 'B' as shown. Pick out the correct output waveform.

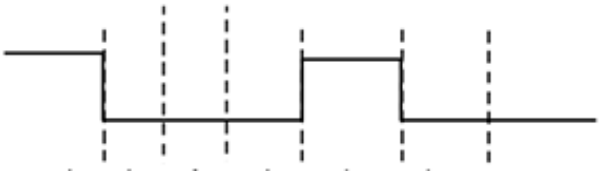
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Output is :

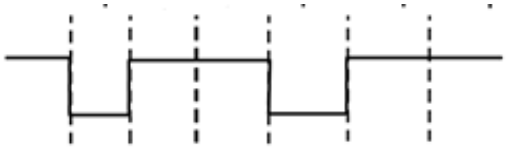




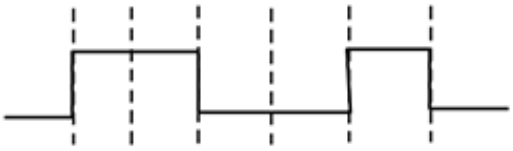
(A)



(B)



(C)

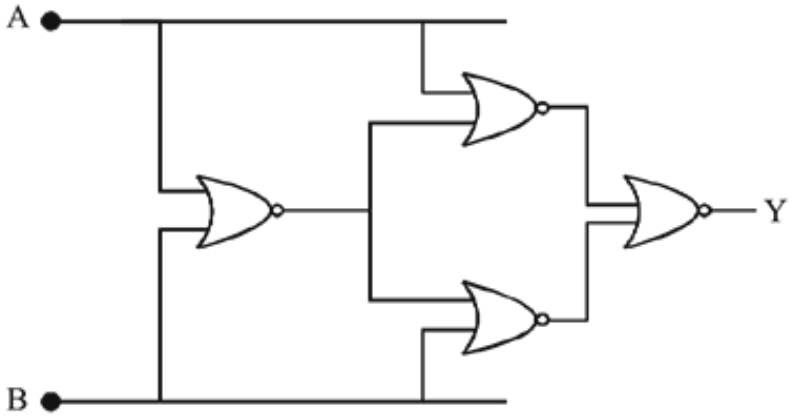


(D)



Q.14

A system of four gates is set up as shown. The 'truth table' corresponding to this system is :



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(A)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(C)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

Q.15

In a full wave rectifier circuit operating from $150\ \text{Hz}$ mains frequency, the fundamental frequency in the ripple would be

(A) $25\ \text{Hz}$

(B) $50\ \text{Hz}$

(C) $70.7\ \text{Hz}$

(D) $300\ \text{Hz}$

Answers & Solutions

Q.1 Answer:

$100\ \text{Hz}$

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Solution:

The output of a full-wave rectifier is of the form

$I = i_0 \left(1 - \cos 2\omega t - \frac{2}{15} \cos 4\omega t - \ldots\right)$

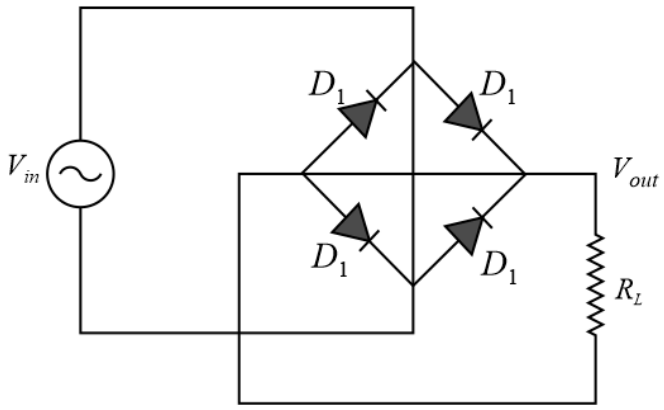
where $i_0 = \frac{2I_o}{\pi}$. The angular frequencies of the harmonic components of the ripple are $2\omega, 4\omega, \ldots$ etc. Thus the fundamental frequency of the ripple is twice the frequency of the mains. Hence the correct choice is (d).

Q.2 Answer:

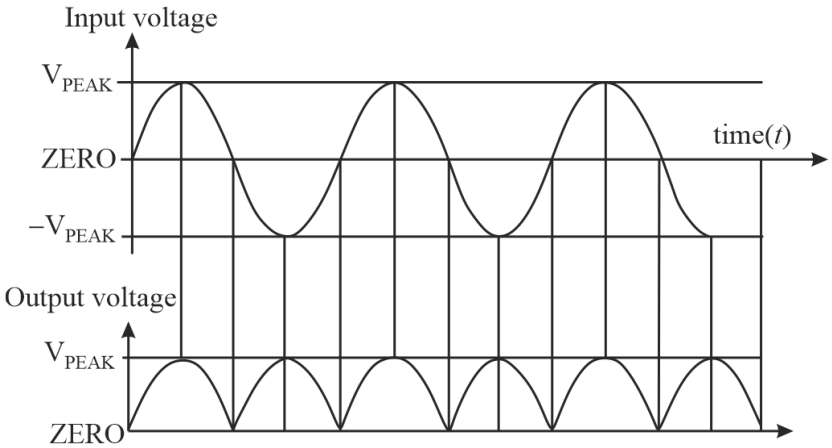
100 Hz

Solution:

A bridge rectifier is basically a full wave rectifier.



Input waveform and output waveform of a full wave rectifier is given below.



We can see from the diagram that for one full wave of input signal, we get two full waves of the output signal. In India, the frequency of AC mains supply is 50 Hz. Therefore, the output frequency of the bridge rectifier will be two times the input frequency, i.e., 100 Hz.

Q.3 Answer:

XOR gate

Solution:

Output of upper AND gate $= \overline{\mathrm{A}B}$

Output of lower AND gate $= \mathrm{AB}$

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$\therefore$ Output of OR gate, $Y = A\overline{B} + B\overline{A}$

This is boolean expression for XOR gate.

Q.4 Answer:

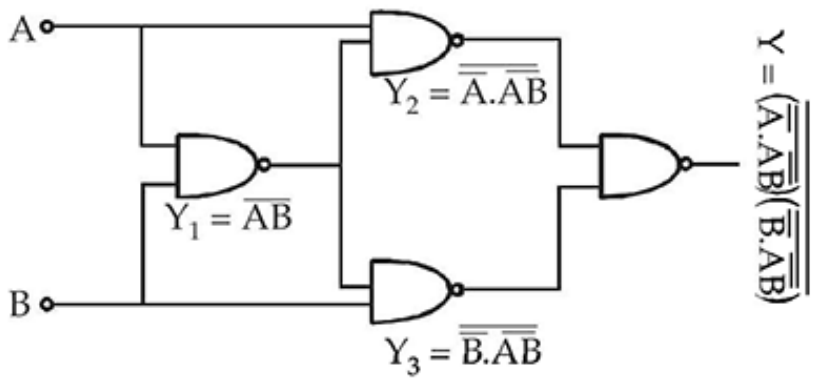
\begin{array}{|l|l|l|}
\hline \mathrm{A} & \mathrm{B} & \mathrm{Y} \\
\hline 0 & 0 & 0 \\
\hline 0 & 1 & 1 \\
\hline 1 & 0 & 1 \\
\hline 1 & 1 & 0 \\
\hline
\end{array}

Solution:

By expanding this Boolen expression

$Y = A \cdot \bar{B} + B \cdot \bar{A}$

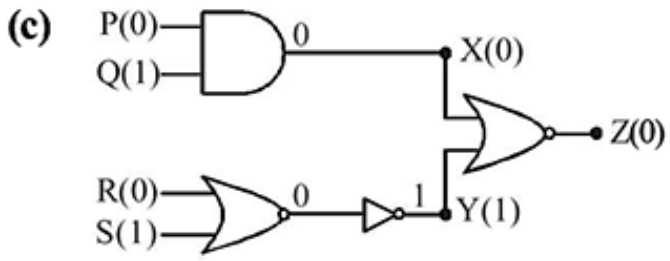
Thus the truth table for this expression should be (a).



Q.5 Answer:

$0, 1, 0$

Solution:



Q.6 Answer:

$W + (X.Y)$

Solution:

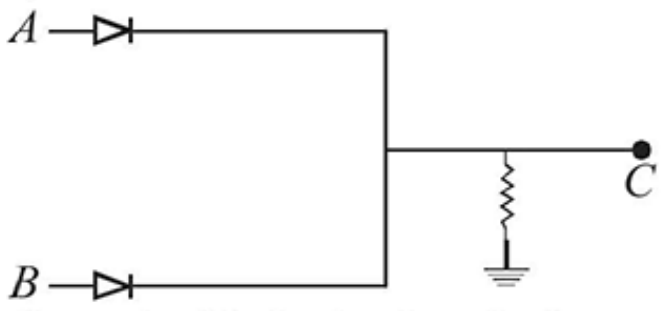
$$(W + X) \cdot (W + Y) = W + (X \cdot Y)$$

Q.7 Answer:

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OR gate

Solution:



The truth table for the above logic gate is :

\begin{array}{|c|c|c|}

**\hline A & B & C **

**\hline 1 & 1 & 1 **

**\hline 1 & 0 & 1 **

**\hline 0 & 1 & 1 **

**\hline 0 & 0 & 0 **

\hline

\end{array} This truth table follows the boolean algebra $C = A + B$ **which is for OR gate**

Q.8 Answer:

$i_2 = 0.25$ A

Solution:

No solution available

Q.9 Answer:

1,1

Solution:

Truth table of the gate is as follows :

\begin{array}{|c|c|c|c|c|}

**\hline A & B & C & A+B & Y=(A+B) \cdot C & \bar{Y} **

**\hline 0 & 0 & 0 & 0 & 0 & 1 **

**\hline 1 & 1 & 0 & 1 & 0 & 1 **

\hline

\end{array}

Q.10 Answer:

XOR gate

Solution:

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Output of upper AND gate $= \overline{AB}$ **Output of lower AND gate** $= A\bar{B}$ $\therefore$ **Output of OR gate,** $Y = A\bar{B} + B\bar{A}$

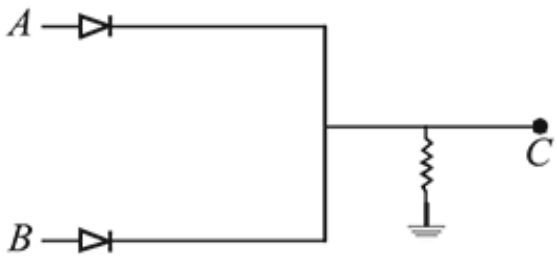
This is boolean expression for XOR gate.

Q.11 Answer:

OR gate

Solution:

The truth table for the above logic gate is :



$A \rightarrow_{1}^{\infty}$ **\begin{array}{|c|c|c|}**

**\hline A & B & C **

**\hline 1 & 1 & 1 **

**\hline 1 & 0 & 1 **

**\hline 0 & 1 & 1 **

**\hline 0 & 0 & 0 **

\hline

\end{array} This truth table follows the boolean algebra $C = A + B$ **which is for OR gate**

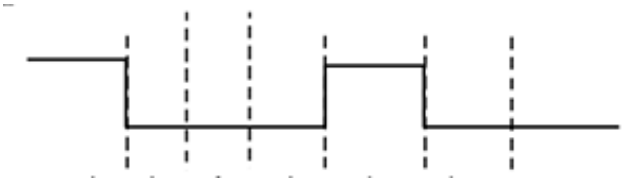
Q.12 Answer:

0,0

Solution:

For $A = 0, B = 0, \overline{A + B} = 1$ **and** $(A \cdot B) = 1$ So $\overline{(A + B)}' \cdot \overline{(A \cdot B)}' = 1$

Q.13 Answer:



Solution:

No solution available

Q.14 Answer:

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\begin{array}{|c|c|c|}
\hline \mathrm{A} & \mathrm{B} & \mathrm{Y} \\
\hline 0 & 0 & 1 \\
\hline 0 & 1 & 0 \\
\hline 1 & 0 & 0 \\
\hline 1 & 1 & 1 \\
\hline
\end{array}

Solution:

No solution available

Q.15 Answer:

300 Hz

Solution:

Input frequency, $f = 150 \text{ Hz} \Rightarrow T = \frac{1}{150}$

For full wave rectifier, $T_1 = \frac{T}{2} = \frac{1}{300} \Rightarrow f_1 = 300 \text{ Hz}$

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